

Emotion Recognition using CNN (FER2013)

Problem Statement

Facial emotion recognition aims to identify human emotions from facial images. This is a challenging real-world image processing problem due to subtle expressions, low image quality, and facial variations.



Algorithm Used – Convolutional Neural Network (CNN)

Why CNN?

- Images are made of pixels, not simple numbers
- CNNs are designed specifically for image data
- They automatically learn important visual features

How CNN Works (3 Stages)

01

Feature Detection

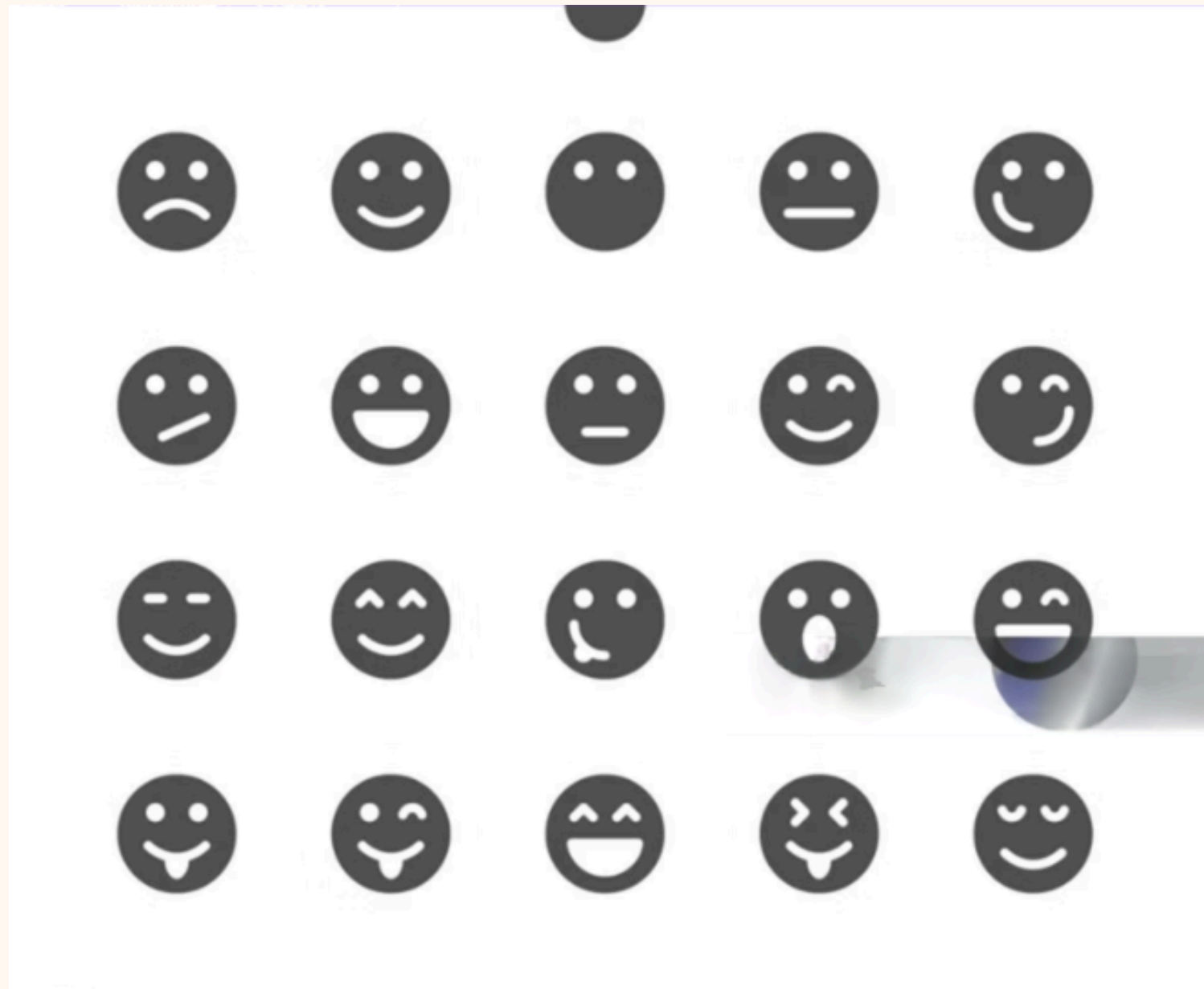
02

Feature Reduction

03

Classification

Dataset Used – FER2013



FER2013 Dataset

- Public facial emotion dataset
- 48×48 grayscale images
- 7 emotion classes
- Collected from real-world conditions

Expected Accuracy

- Emotion recognition is a difficult task
- 50–60% accuracy is considered reasonable for FER2013

Methodology & Key Learnings

Methodology

- Image preprocessing and normalization
- Data augmentation to improve learning
- CNN model training
- Emotion prediction on new images

Key Learnings

- How CNNs process images step by step
- Importance of data quality and preprocessing
- Real-world image tasks are challenging